

4.1
 $a_1, N=20, \sum y_i = 7825,94, \bar{y} = 19,21, SSR = 375,47$

$$SST = \sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N y_i^2 - \sum_{i=1}^N 2 y_i \bar{y} + n \bar{y}^2 = 7825,94 - 2 \times 19,21 \times 20 \times 19,21 + 20 \times 19,21^2$$

$$= 7825,94 - 20 \times 19,21^2$$

$$= 445,458$$

$$\Rightarrow R^2 = \frac{SSR}{SST} = \frac{375,47}{445,458} = 0,84288 \div 0,843 \#$$

b, $R^2 = 0,7911, SST = 725,94, N = 20 \left[R^2 = 1 - \frac{SSE}{SST} \Rightarrow SSE = (1 - R^2) \times SST \right]$

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N-2} = \frac{SSE}{N-2} = \frac{SST \times (1 - R^2)}{N-2} = \frac{725,94 \times (1 - 0,7911)}{18} = 8,4249 \#$$

c, $\sum_{i=1}^n (y_i - \bar{y})^2 = 631,63, \sum \hat{e}_i^2 = 182,85$

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{182,85}{631,63} = 0,71051 \div 0,711 \#$$

4.3

a, $\hat{y}_0 = 1,2 + 0,8x_0 = 1,2 + 0,8 \times 4 = 4,4 \#$

b,

$$se(f) = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]} = \sqrt{\frac{\sum \hat{e}_i^2}{N-2} \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]}$$

$$= \sqrt{\frac{3,6}{3} \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right]} = 1,58745 \div 1,5875 \#$$

c, 95% PI for $x_0 = 4$ ($\hat{y}_0 = 4,4$) is

$$P(\hat{y}_0 - t_c \times se(f) < y_0 < \hat{y}_0 + t_c \times se(f)) \quad , \quad t_c = t_{3,0,025} = 3,182, se(f) = 1,5875$$

$\therefore$ the 95% PI for y_0 is $[4,4 - 1,5875 \times 3,182, 4,4 + 1,5875 \times 3,182]$

$$= [-0,6514, 9,4514] \#$$

D, change t_c to $t_{3,0,005} = 5,841$

$$\text{and the 99\% PI for } y_0 \text{ is } [4,4 - 1,5875 \times 5,841, 4,4 + 1,5875 \times 5,841] \#$$

$$= [-4,8726, 13,6726]$$

續 4.3

e, the 95% PI for y given $\bar{x}$ is $[\hat{y} - t_{3,0.025} \times S \sqrt{1 + \frac{1}{5}}, \hat{y} + t_{3,0.025} \times S \sqrt{1 + \frac{1}{5}}]$

$$\hat{y} = 1.2 + 0.8 \times 1 = 2$$
$$= [2 - 3.182 \times \sqrt{\frac{3.6}{3}} \times \sqrt{\frac{6}{5}}, 2 + 3.182 \times \sqrt{\frac{3.6}{3}} \times \sqrt{\frac{6}{5}}]$$

$$= [-1.819, 5.819]$$

$$\begin{cases} \text{width for (c)} = 9.4514 - (-0.6514) = 10.1028 \end{cases}$$

$$\begin{cases} \text{width for (e)} = 5.819 - (-1.819) = 7.638 \end{cases}$$

(e) is shorter since $\frac{(x - \bar{x})^2}{s} = 0$

4,25

(a)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	4.393866	0.043288	101.50	<2e-16 ***
## sqft	0.036044	0.001486	24.26	<2e-16 ***

(a)

slope at the sample mean is : 9.019661

elasticity at the sample mean is : 0.9833701

b₁:when $\ln(\text{sqft}) = 0$, $\ln(\widehat{\text{price}}) \div 4.3939$

$$\Rightarrow \widehat{\text{price}} \div e^{4.3939} = 80.9$$

b₂:

其他條件不變, sqft 增加1單位

→ 房價約增加 3.6%

(b)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	2.04965	0.15797	12.97	<2e-16 ***
## log(sqft)	1.02483	0.04839	21.18	<2e-16 ***

(b)

slope at the sample mean is : 9.399923

elasticity at the sample mean is : 1.024828

 α_1 : when $\ln(\text{sqft}) = 0$ ($\text{sqft} = 1$)

$$\Rightarrow \ln(\widehat{\text{price}}) \div 2.0497$$

$$\Rightarrow \widehat{\text{price}} \div e^{2.0497}$$

 α_2 : sqft 增加1%, price 增加 1.0248%

(c)

(c)

R-squared value from the linear model is : 0.6413167

Generalized R-squared from (b) is : 0.6445084

Generalized R-squared from (c)(the linear model) is : 0.6413167

model (b) better

than model (c)

(d)

(d)

the Jarque-Bera statistics for (a) is : 26.67899

the Jarque-Bera statistics for (b) is : 14.27871

the Jarque-Bera statistics for (c) is : 221.1115

(d)

p-value for (a) is : 1.609651e-06

p-value for (b) is : 0.0007932632

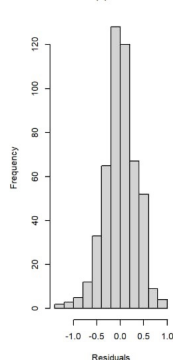
p-value for (c) is : 0

3個 model 的 p-value 皆很小

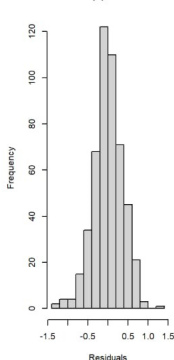
⇒ 殘差的分配不完全符合常態分佈

model (2) 的 JB-statistic 最小 ⇒ 相對較接近

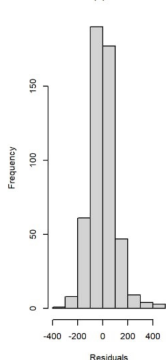
Model (a) Residuals



Model (b) Residuals

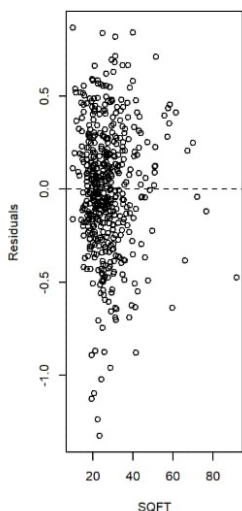


Model (c) Residuals

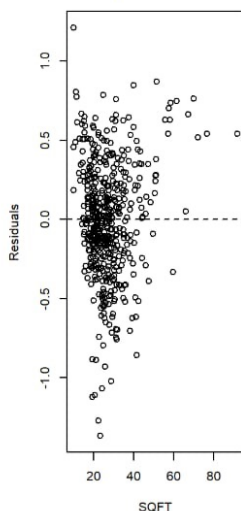


(e)

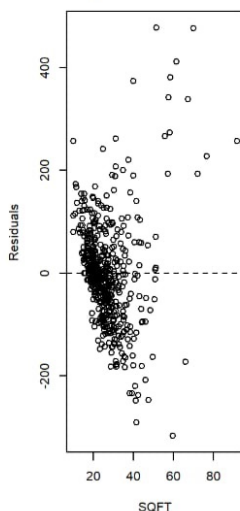
Model (a): Residuals vs SQFT



Model (b): Residuals vs SQFT



Model (c): Residuals vs SQFT



殘差分佈合理

(f), (g)

```
# (f)
# predict value for model (a) is : 214233.6
# predict value for model (b) is : 227538.6
# predict value for model (c) is : 246455.7
```

(g)

```
## (X): 95%PI
## model (a) log-linear: 109.7685,418.1167
## model (b) log-log: 111.1406,465.8406
## model (c) linear : 44.27727,448.6341
```

(h)

linear model 表現較不好, 不適合用. (JB-statistic too large)

log-linear 和 log-log 之間 前者 R^2 較高, 後者 JB-statistic 較小

兩者都可選用。

5,5

$$(a) \text{Var}(\hat{\beta}_2 | X) = \text{Var}(b_2 | X) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} = \frac{\sigma^2}{\sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T}}$$

$$\begin{aligned} \therefore \sum_{t=1}^T x_t^2 - \frac{(\sum_{t=1}^T x_t)^2}{T} & \stackrel{x_t = t, t=1, \dots, T}{=} \sum_{t=1}^T t^2 - \frac{(\sum_{t=1}^T t)^2}{T} = \frac{T(T+1)(2T+1)}{6} - \frac{(\sum_{t=1}^T t)^2}{T} \\ & = \frac{T(T+1)(2T+1)}{6} - \frac{1}{T} \times \left(\frac{T(T+1)}{2}\right)^2 \\ & = \frac{T(T+1)(2T+1)}{6} - \frac{T(T+1)^2}{4} \\ & = \frac{2T(T+1)(2T+1) - 3T(T+1)^2}{12} \\ & = \frac{4T^3 + 6T^2 + 2T - 3T^3 - 6T^2 - 3T}{12} \\ & = \frac{T^3 - T}{12} \end{aligned}$$

$$\Rightarrow \text{Var}(b_2 | X) = \frac{12\sigma^2}{T(T+1)(T-1)} \quad \text{and} \quad \lim_{T \rightarrow \infty} \frac{12\sigma^2}{T(T+1)(T-1)} = 0 \quad \therefore \text{the LSE estimator } b_2 \text{ is consistent}$$

$$(b) \text{Var}(\hat{\beta}_2 | X) = \text{Var}(b_2 | X) = \frac{\sigma^2}{\sum_{t=1}^T (x_t - \bar{x})^2} = \frac{\sigma^2}{\sum_{t=1}^T (0,5^t - \bar{x})^2}$$

$$\begin{aligned} \sum_{t=1}^T (0,5^t - \bar{x})^2 & = \sum_{t=1}^T 0,5^{2t} - 2 \times \bar{x} \times \sum_{t=1}^T 0,5^t + T \bar{x}^2 = \sum_{t=1}^T 0,5^{2t} - T \bar{x}^2 = \sum_{t=1}^T 0,25^t - \frac{1}{T} \left(\sum_{t=1}^T 0,5^t\right)^2 \\ & = \frac{0,25(1-0,25^T)}{1-0,25} - \frac{1}{T} \times \frac{0,5(1-0,5^T)}{1-0,5} \\ & = \frac{1}{3}(1-0,25^T) - \frac{1}{T}(1-0,5^T) \end{aligned}$$

$$\Rightarrow \text{Var}(b_2 | X) = \frac{\sigma^2}{\frac{1}{3}(1-0,25^T) - \frac{1}{T}(1-0,5^T)}$$

$$\text{and} \quad \lim_{T \rightarrow \infty} \text{Var}(b_2 | X) = \frac{\sigma^2}{\frac{1}{3} \times 1 - 0} = 3\sigma^2 \neq 0 \quad \therefore \text{the LSE estimator } b_2 \text{ isn't consistent}$$

(c) in (a), increasing T provides more information about β_2 and the relationship between y and x . however, in (b), $x = 0,5^t$ become 0 for moderately sized T , and increasing T further does not provide any more information about β_2

5,6 $N = 63$, $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$, $\widehat{\text{Cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$

(a) $H_0: \beta_2 = 0$ v.s. $H_1: \beta_2 \neq 0$

$\therefore \frac{b_2 - 0}{\text{se}(b_2)} \sim t_{60, 0.025} = 2.000 \Rightarrow \text{reject } H_0 \text{ if } |t^*| > 2$

$t^* = \frac{b_2}{\text{se}(b_2)} = \frac{3}{\sqrt{4}}$, since $|t^*| = \frac{3}{2} < 2$, $\therefore$ do not reject the null hypothesis.

(b) $H_0: \beta_1 + 2\beta_2 = 5$ v.s. $\beta_1 + 2\beta_2 \neq 5$

$\therefore \frac{(b_1 + 2b_2) - 5}{\text{se}(b_1 + 2b_2)} \sim t_{60, 0.025} = 2 \Rightarrow \text{reject } H_0 \text{ if } |t^*| > 2$

$t^* = \frac{(b_1 + 2b_2) - 5}{\text{se}(b_1 + 2b_2)} = \frac{2 + 3 \times 2 - 5}{\sqrt{3 + 4 \times 4 + 2 \times 2 \times (-2)}} = \frac{3}{\sqrt{11}} < 2$, $\therefore$ do not reject the null hypothesis

(c) $H_0: \beta_1 - \beta_2 + \beta_3 = 4$ v.s. $H_1: \beta_1 - \beta_2 + \beta_3 \neq 4$

$\therefore \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} \sim t_{60, 0.025} = 2 \Rightarrow \text{reject } H_0 \text{ if } |t^*| > 2$

$t^* = \frac{(b_1 - b_2 + b_3) - 4}{\text{se}(b_1 - b_2 + b_3)} = \frac{2 - 3 + (-1) - 4}{\sqrt{3 + (-1)^2 \times 4 + 3 + 2 \times (-1) \times (-2) + 2 \times 0 + 2 \times 1}} = \frac{-6}{\sqrt{16}}$

since $|t^*| = \frac{6}{4} < 2$, $\therefore$ do not reject the null hypothesis

5, > 0

TABLE 2.4

Exercise 2.3 Data

x	1	2	3	4	5	6
y	6	4	11	9	13	17

$$(a) \bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \frac{1}{3} \sum_{i=1}^3 e_i, \quad \bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \frac{1}{3} \sum_{i=4}^6 e_i, \quad \text{let } \frac{1}{3} \sum_{i=1}^3 e_i = \bar{e}_1, \quad \frac{1}{3} \sum_{i=4}^6 e_i = \bar{e}_2$$

$$\Rightarrow \hat{\beta}_{2, \text{mean}} = \frac{\bar{y}_1 - \bar{y}_2}{\bar{x}_1 - \bar{x}_2} = \beta_2 + \frac{\bar{e}_1 - \bar{e}_2}{\bar{x}_1 - \bar{x}_2}$$

$$\therefore E(\hat{\beta}_{2, \text{mean}} | x) = \beta_2 + \frac{1}{\bar{x}_2 - \bar{x}_1} (E(\bar{e}_1 - \bar{e}_2 | x)) = 0 \quad (\because E(\bar{e}_i | x) = 0, i=1,2)$$

$$\Rightarrow = \beta_2 + \frac{1}{\bar{x}_2 - \bar{x}_1} \times 0 = \beta_2$$

$$\therefore E(\hat{\beta}_{2, \text{mean}}) = E(E(\hat{\beta}_{2, \text{mean}} | x)) = E(\beta_2) = \beta_2 \quad \text{Hence } \hat{\beta}_{2, \text{mean}} \text{ is an unbiased estimator of } \beta_2$$

(b)

$$\therefore \hat{\beta}_{2, \text{mean}} - \beta_2 = \frac{\bar{e}_1 - \bar{e}_2}{\bar{x}_1 - \bar{x}_2} \Rightarrow \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \frac{\text{Var}(\bar{e}_2 - \bar{e}_1 | x)}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$\text{note that } \text{Var}(\bar{e}_2 - \bar{e}_1 | x) = \text{Var}(\bar{e}_2 | x) + \text{Var}(\bar{e}_1 | x)$$

$$= \text{Var}\left(\frac{1}{3} \sum_{i=4}^6 e_i \mid x\right) + \text{Var}\left(\frac{1}{3} \sum_{i=1}^3 e_i \mid x\right)$$

$$= \frac{1}{9} \times 3 \sigma^2 + \frac{1}{9} \times 3 \sigma^2 = \frac{2\sigma^2}{3}$$

$$\therefore \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \frac{\frac{2\sigma^2}{3}}{3(\bar{x}_2 - \bar{x}_1)^2} \quad \#$$

(c)

$$\text{note that: } \text{Var}(\hat{\beta}_{2,\text{mean}}) = E[\text{Var}(\hat{\beta}_{2,\text{mean}} | \chi)] + \text{Var}(E(\hat{\beta}_{2,\text{mean}} | \chi))$$

$$\begin{cases} \text{Var}(E(\hat{\beta}_{2,\text{mean}} | \chi)) = \text{Var}(\beta_2) = 0 \end{cases}$$

$$\begin{cases} E(\text{Var}(\hat{\beta}_{2,\text{mean}} | \chi)) = E\left[\frac{2\sigma^2}{3(\bar{x}_2 - \bar{x}_1)^2}\right] = \frac{2\sigma^2}{3} E\left[\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right] \end{cases}$$

$$\therefore \text{Var}(\hat{\beta}_{2,\text{mean}}) = \frac{2\sigma^2}{3} \times E\left[\frac{1}{\bar{x}_2 - \bar{x}_1}\right]$$

(d)

$$\text{Var}(\hat{\beta}_{2,\text{mean}} | \chi) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} = \frac{4\sigma^2}{9N} \Rightarrow \text{Var}(\hat{\beta}_{2,\text{mean}} | \chi) \rightarrow 0 \text{ as } N(\bar{x}_2 - \bar{x}_1)^2 \rightarrow \infty$$

$$\text{Hence } \text{Var}(\hat{\beta}_{2,\text{mean}} | \chi) \text{ is constant if } \lim_{N \rightarrow \infty} N(\bar{x}_1 - \bar{x}_2)^2 = \infty$$

(e), (f) 放後面

(g)

$$\textcircled{1} E[(S_X^2)^{-1}] \xrightarrow{P} 0.12$$

$$\because X_i \sim U(0, 10), \text{ i.i.d. and } \text{Var}(X) = \frac{(10-0)^2}{12} = \frac{100}{12}$$

$$\text{by 大數法則: } S_X^2 = \frac{1}{N-1} \left(\sum_{i=1}^N (X_i - \bar{X})^2 \right) \xrightarrow{P} \text{Var}(X) = \frac{100}{12}$$

$$\therefore S_X^{-2} \xrightarrow{P} \frac{100}{12} \Rightarrow S_X^{-2} \xrightarrow{P} \left(\frac{100}{12}\right)^{-1} = 0.12$$

$$\text{and Hence } E(S_X^{-2}) \xrightarrow{P} 0.12$$

$$\textcircled{2} \frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{P} 0.16$$

note that 前半樣本集中於 (0, 5), 而後半樣本集中於 (5, 10)

$$\text{and } E(X | X \in (0, 5)) = \frac{5+0}{2} = 2.5, E(X | X \in (5, 10)) = \frac{5+10}{2} = 7.5$$

$$\text{Hence when } N \rightarrow \infty \Rightarrow \bar{x}_1 \xrightarrow{P} 2.5, \bar{x}_2 \xrightarrow{P} 7.5 \Rightarrow \bar{x}_2 - \bar{x}_1 \xrightarrow{P} 5$$

$$\text{and } \frac{4}{(\bar{x}_2 - \bar{x}_1)^2} \xrightarrow{P} \frac{4}{5^2} = 0.16, \text{ Hence } E\left(\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right) = 0.16$$

(h)

$\hat{\beta}_{2, \text{mean}}$ will not be consistent.

$\therefore$ for $\bar{x}_1, \bar{x}_2$, 都會 converge in probability to 5 (都是 random sample)

$$\Rightarrow \bar{x}_2 - \bar{x}_1 \xrightarrow{p} 0$$

$$\therefore \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \text{ 無法確定是否收斂到 } 0$$

(e) (i) (ii)

	$\text{Var}(b_2 x)$	$\text{Var}(\hat{\beta}_{2, \text{mean}} x)$	$E[Sx^{-2}]$	$E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right]$
1 100	1.24362517	1.6631963	0.1243625	0.1663196
2 500	0.24402399	0.3260404	0.1220120	0.1630202
3 1000	0.12290628	0.1656282	0.1229063	0.1656282
4 5000	0.02413174	0.0323161	0.1206587	0.1615805

(f)

$\textcircled{1}$ by the result of (e),
$$\begin{cases} \text{Var}(b_2 | x) = \frac{\sigma^2}{N S x^{-2}}, & \text{Var}(\hat{\beta}_{2, \text{mean}} | x) = \frac{4\sigma^2}{N(\bar{x}_2 - \bar{x}_1)^2} \\ E(Sx^{-2}) \doteq 0.12, & E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \doteq 0.16 \end{cases}$$

$$\text{Hence } E[\text{Var}(b_2 | x)] = \frac{\sigma^2}{N} E(Sx^{-2}) \approx \frac{0.12\sigma^2}{N}$$

$$\text{and } E[\text{Var}(\hat{\beta}_{2, \text{mean}} | x)] = \frac{\sigma^2}{N} E\left[\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right] \approx \frac{0.16\sigma^2}{N}$$

$$\therefore \text{Var}(\hat{\beta}_{2, \text{mean}}) > \text{Var}(b_2)$$

$\textcircled{2}$ 當 N 越大, 兩者分別越下降且趨近於 0.12 跟 0.16

$\textcircled{3}$ $\therefore x_i \sim U(0, 10) \Rightarrow \text{when } N \rightarrow \infty, \bar{x}_1 \rightarrow 2.5, \bar{x}_2 \rightarrow 7.5, \therefore \text{Var}(\hat{\beta}_{2, \text{mean}} | x) \rightarrow \frac{4\sigma^2}{(7.5 - 2.5)^2 N}$

$$= \frac{0.16\sigma^2}{N} \rightarrow 0 \text{ as } N \rightarrow \infty$$